

AutoNexa

Autonomous Parking and Recall System

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Low-cost indoor autonomy with live LiDAR mapping, Ackermann navigation, mobile control, and layered safety

Project Description

AutoNexa is a low-cost autonomous parking and recall prototype for structured indoor environments. It builds a live LiDAR map, localizes itself, navigates to user-selected targets, parks in marked spaces, and returns to a recall point through the mobile application.

Map-free navigation

Uses 2D LiDAR scans and live SLAM instead of a pre-saved map.

Integrated autonomy stack

Combines LiDAR mapping, scan-matcher odometry, Nav2 planning, and Pico-based actuation.

Vehicle-like motion

Uses Ackermann steering for realistic forward and reverse parking maneuvers.

Obstacle awareness

Adds LiDAR-detected obstacles to the costmap for collision-aware navigation.

Mobile operation

Allows manual driving, goal selection, waypoint storage, monitoring, and E-STOP from a phone.

Layered safety

Filters planner commands through software and firmware safety layers before motor actuation.

Solution

AutoNexa is built as a layered autonomy pipeline from sensing to actuation. LiDAR data is used to create the map and pose estimate, Nav2 generates feasible vehicle-like paths, and the Pico converts safe velocity commands into steering and motor outputs.

Perception

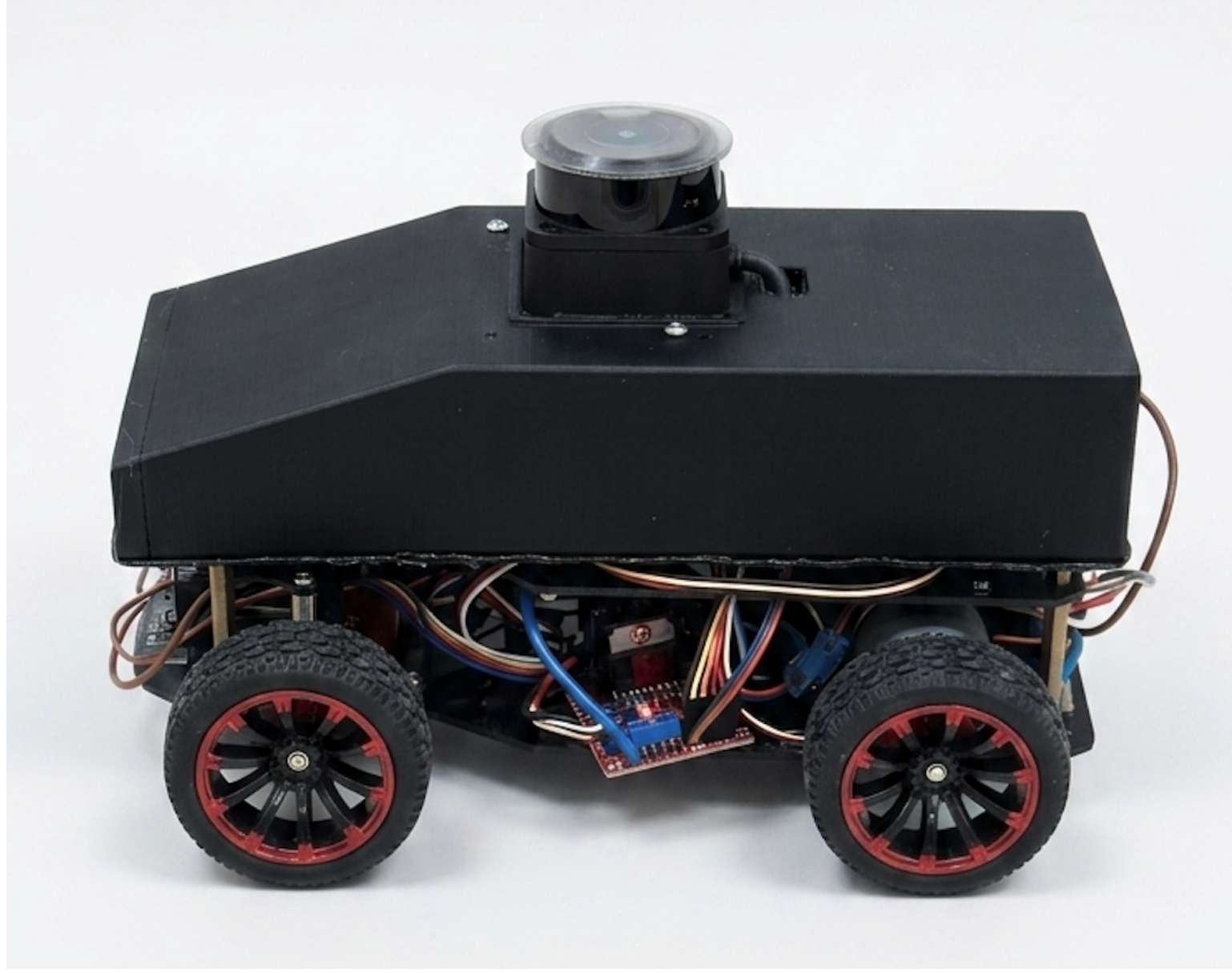
SLAMTEC C1 LiDAR provides 360° scans. The scan filter chain, SLAM Toolbox, and laser scan matcher produce the live map and odometry.

Navigation Control

SMAC Hybrid-A* plans Ackermann-compatible paths, while Regulated Pure Pursuit tracks the path at low speed. The Pico drives the servo and rear motors.

Mobile Interface

The mobile app shows the live map, stores parking and recall waypoints, sends goals, supports manual driving, and triggers E-STOP.



Strengths & Differentiators

No markers. No pre-mapping. No instrumented parking spots.

Typical parking demos tag *every* spot with an ArUco code and keep a camera locked on it — so they only work in that one prepared layout. AutoNexa needs none of that and drops into *any* unprepared indoor space.

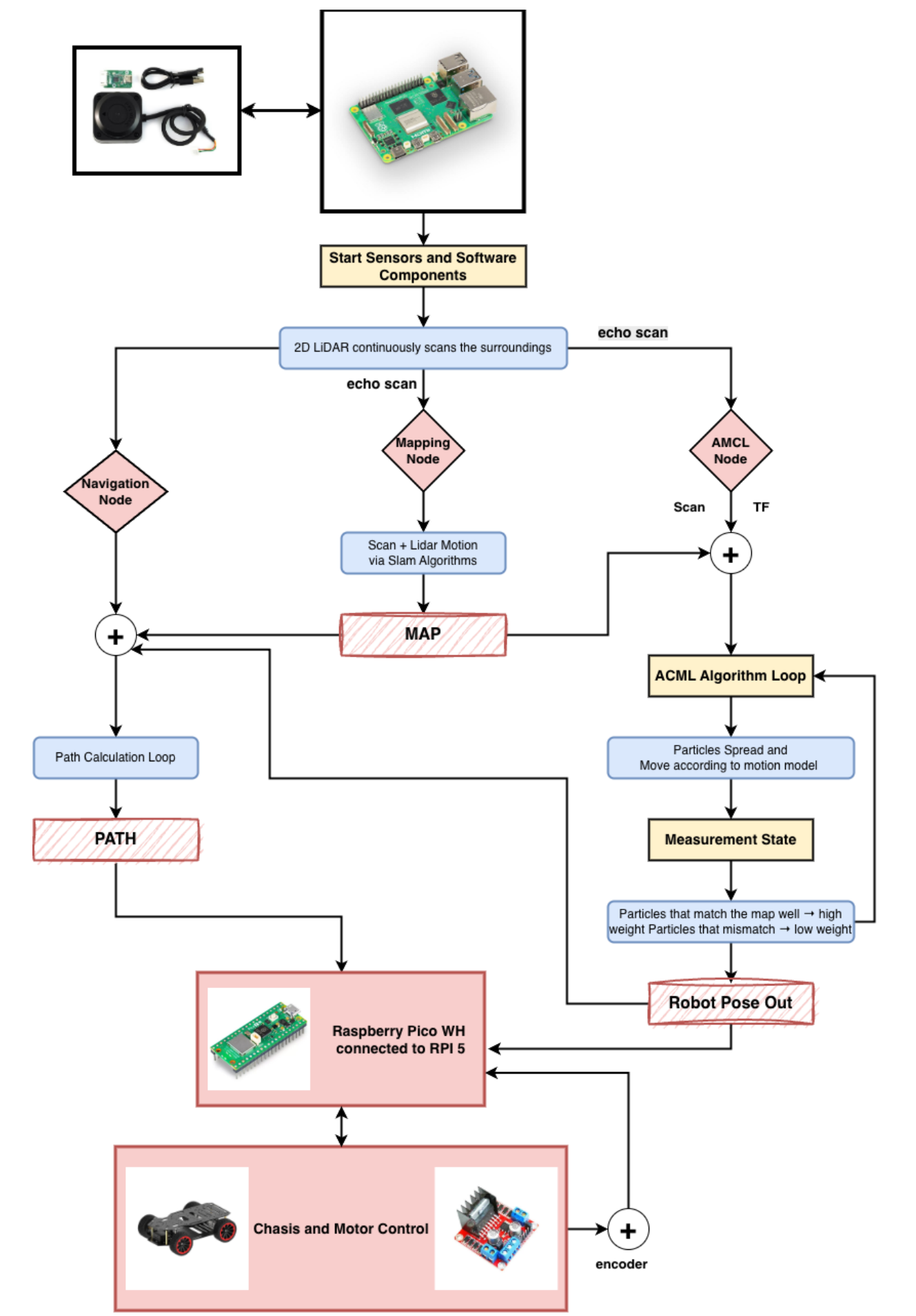
Headless phone-only operator console: map, live tuning, waypoints, and calibration — no monitor or RViz on site.

Ackermann reverse-capable planning for true car-like parking, not turn-in-place.

Cost Analysis

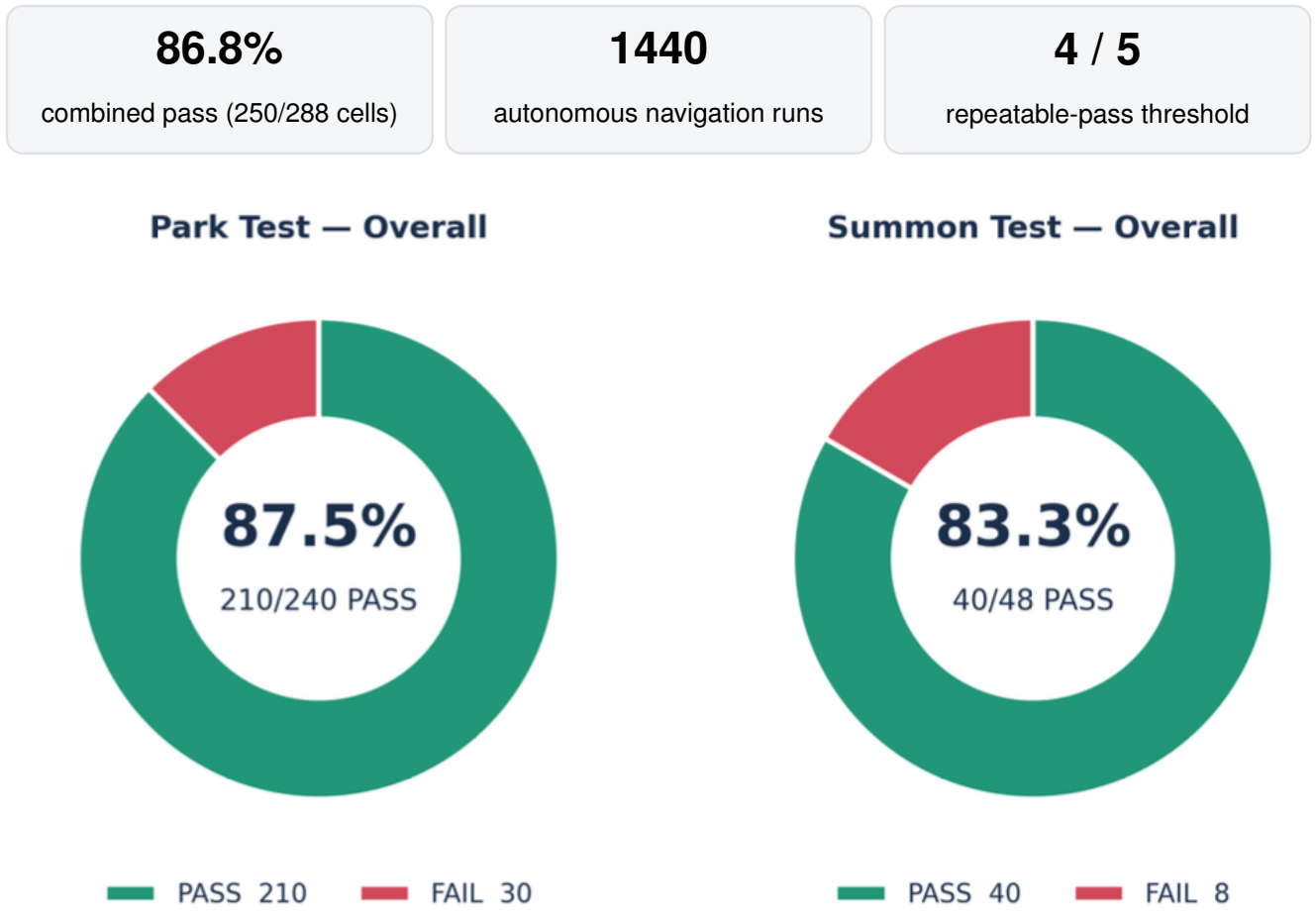
Product	Cost
Steering servo	\$4.36
Ackermann chassis + DC motors	\$23.99
Power system	\$37.71
L298N motor driver	\$3.27
Raspberry Pi 5 + Raspberry Pi Pico	\$88.00
SLAMTEC C1 LiDAR	\$105.00
Total product cost	\$262.34
Including mounts, cables, replacement parts, and shipping, the complete hardware and infrastructure cost is \$297.34 .	

Design

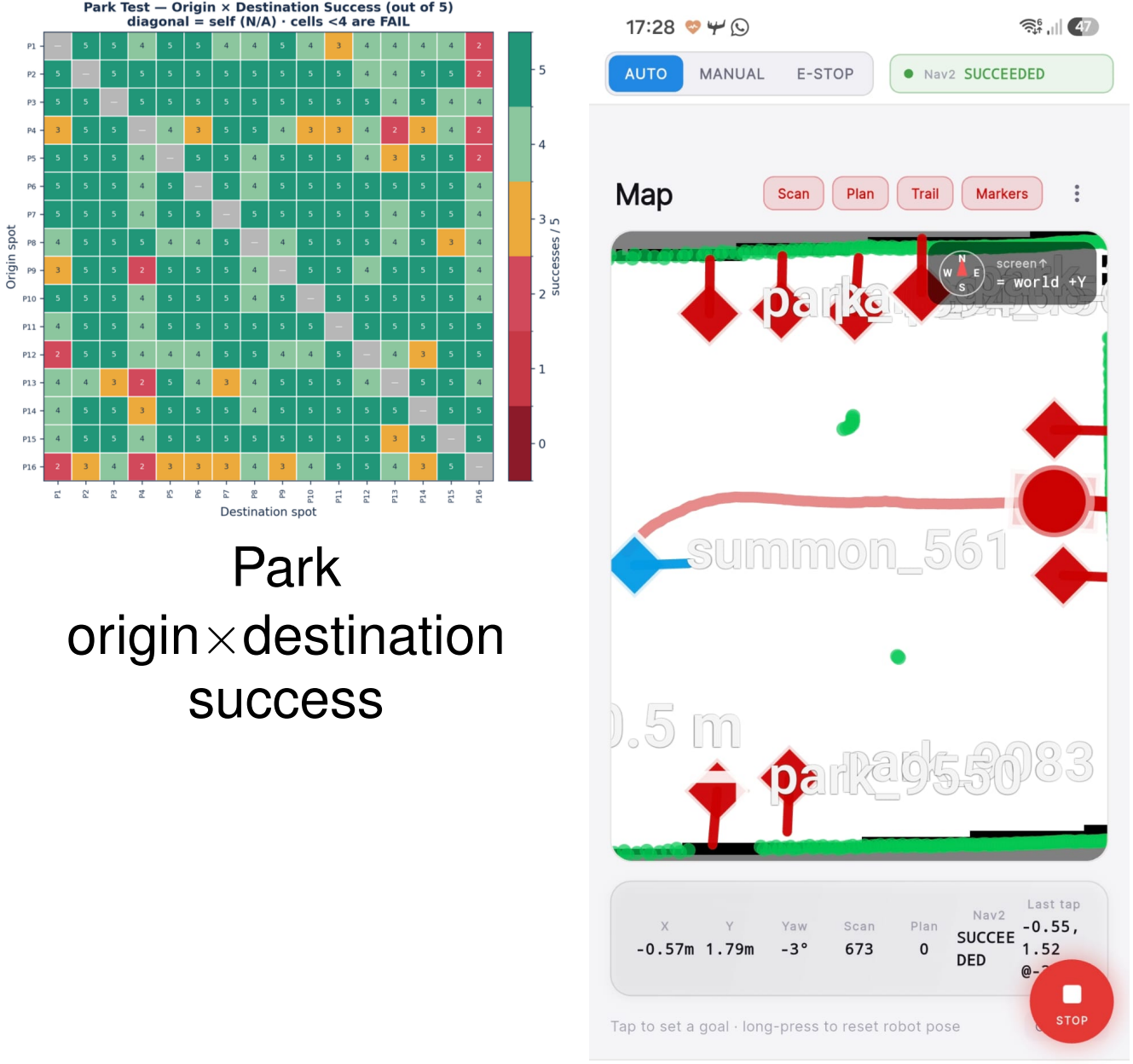


AutoNexa data-flow architecture

Test Results



Overall pass rates — park 87.5% (210/240), summon 83.3% (40/48)



Live mobile run

Deliverables

Autonomous Vehicle

Self-parking & recall robot

Mobile App

Goals, driving & E-STOP

ROS 2 Stack

SLAM, Nav2 & firmware

Documentation

Final report & test results